Heart disease prevalence

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First Name: **Last Name**:

Instructions

The exam duration is **one hour and a half**.

For this exam, **all documents are allowed** (including all previous notebooks), fact sheets/information sheets you may have prepared.

Your answers must be complete but there is no need to give the code you typed unless otherwise specified. Show your work, including formulas and calculations where appropriate. Interpret all results in the context of Alzheimer's disease research and potential clinical implications.

Note about α threshold: For all statistical tests, use a **significance level of $\alpha = 0.03$**.

Dataset

You are analyzing data from a heart health study conducted on 150 adults aged 25-65. The dataset `heart_health.csv` contains the following variables:

- `patient_id`: Unique patient identifier
- `age`: Age in years
- `gender`: Gender (Male/Female)
- `bmi`: Body Mass Index (kg/m^2)
- `systolic_bp`: Systolic blood pressure (mmHg)
- `diastolic_bp`: Diastolic blood pressure (mmHg)
- `cholesterol`: Total cholesterol level (mg/dL)
- `exercise_freq`: Exercise frequency per week (0-7 days)
- `smoking_status`: Smoking status (Never/Former/Current)
- `heart_disease`: Presence of heart disease (Yes/No)

Questions

Load the dataset using: `heart_data <- read.csv("heart_health.csv")`

Describing the dataset

Question 1. (1 point) **No need to give the R code**
How many patients are below 35 years old?

Question 2. (2 points) **No need to give the R code**
What is the average BMI of individuals aged over 60?

Question 3. (3 points) **Give the R code**
Create a boxplot showing the distribution of systolic blood pressure by gender. Include appropriate labels and title, and interpret the plot in 2-3 sentences discussing any visible differences between groups.

Testing simple hypotheses

Question 4. (3 points)
Test whether there is a difference in mean BMI between patients with and without heart disease. State your hypotheses, perform the test, **give the code you used to perform the test**, and interpret your findings at = 0.03.

Question 5. (4 points)
Investigate whether there is an association between smoking status and heart disease. Perform the appropriate test, give its hypotheses, and interpret your results. What can you conclude about the relationship between smoking and heart disease?

Modeling cholesterol levels

Question 7. (7 points)
Build a linear regression model to predict cholesterol levels using age and BMI as predictors.

a) **Give the R code you use to fit the model** (1 point)

b) Report the regression equation with coefficient values (2 points)

c) Interpret the coefficient for age in the context of the study (2 points)

d) Test the overall significance of the model and report the R^2 value, what can you say about the model? (2 points)

Bonus Questions

(1 point each, not counted in the main 20-point total)

These questions should be directly answered on the answer sheet!

Bonus Question 1. (1 point)
In a clinical trial comparing two treatments, why might researchers choose a paired t-test instead of a two-sample t-test?

Bonus Question 2. (1 point)
A researcher finds a correlation coefficient of r = 0.85 between two variables and concludes that one variable causes the other. What is wrong with this conclusion? Name one alternative explanation for this strong correlation.